

Qryptic Inc. Polynomial Multiplier IP

Highlights

- Polynomial multiplication for CRYSTALS-Kyber / ML-KEM, degree 256, modulus 3329
- The same ring covers ML-KEM-512, ML-KEM-768, and ML-KEM-1024
- Degree and modulus set per delivery, for other algorithms
- Butterfly units per NTT stage, chosen by the customer as 1, 2, 4, or 8. Doubling this doubles the throughput
- One Kyber polynomial multiplication takes 128 cycles at 1 butterfly unit per stage, 64 at 2, 32 at 4, and 16 at 8. This is a very low number
- Zero block RAM
- Each stage starts when the coefficients it needs are ready
- Twiddle factors computed as data moves; no separate bit-reversal step
- Choice to transform the second polynomial in the hardware, or not
- Streaming interface with backpressure
- Kyber products checked against an independent schoolbook multiplier; code coverage collected

Applications

- ML-KEM key generation, encapsulation, and decapsulation
- Other polynomial multiplies once degree and modulus are set
- Hardware security modules
- FPGA security subsystems
- ASIC multiply blocks

Technology

- Measured on Xilinx Artix-7 xc7a100tftg256-1L
- Place-and-route, out of context
- Delivered for an FPGA device, or ported to an ASIC, as scoped

Overview

The Qryptic polynomial multiplier IP streams two polynomials through a number-theoretic transform, a pairwise multiply, an inverse transform, and a final scale, and streams the product out. The build that has been implemented and measured is CRYSTALS-Kyber / ML-KEM: degree 256, modulus 3329.

Degree and modulus are chosen for each delivery. A different degree changes the number of transform stages. A different modulus changes the coefficient width and the modular reduction. Separately, the customer sets how many butterfly units sit in each NTT stage. The choices are 1, 2, 4, and 8 units per stage. Doubling the units per stage doubles the throughput, and it raises the lookup-table, flip-flop, and DSP cost. On the Kyber build one polynomial multiplication takes 128 cycles at 1 unit per stage, 64 at 2, 32 at 4, and 16 at 8. This is a very low number.

On the Kyber build the sequence is the FIPS 203 transform, and the final scale is 3303, which is $128^{-1} \bmod 3329$.

Each stage starts as soon as the coefficients it needs are ready, and keeps only the alignment storage for that stage. Twiddle factors are computed in step with the data. The schedule emits coefficients in the order the multiply consumes, so there is no separate bit-reversal step. Every measured build uses zero block RAM; the alignment state is in registers and distributed memory.

You choose whether the second polynomial is transformed in the hardware. The interface supports backpressure.

Block diagram

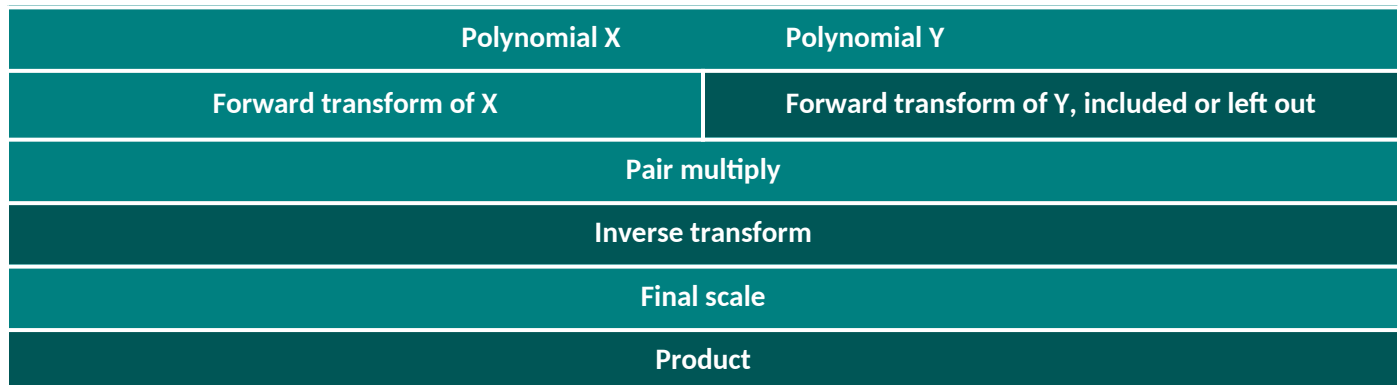


Figure 1. Cryptic polynomial multiplier.

Key features

- Streaming multiply of two polynomials, one coefficient beat per clock when there is no backpressure
- Kyber arithmetic is the FIPS 203 transform: forward transform, pairwise multiply, inverse transform, final scale by 3303
- ML-KEM-512, ML-KEM-768, and ML-KEM-1024 share degree 256 and modulus 3329, so one Kyber delivery covers all three
- Degree, set per delivery: a longer polynomial adds transform stages
- Modulus, set per delivery: coefficient width and the reduction follow the modulus. The Kyber build uses 12-bit coefficients
- Butterfly units per NTT stage: a customer setting of 1, 2, 4, or 8. Doubling it doubles the throughput
 - Each beat carries $2 \times$ (units per stage) coefficients, paired from the low half and the high half of the polynomial
 - One Kyber polynomial multiplication takes 128 cycles at 1 unit per stage, 64 at 2, 32 at 4, and 16 at 8. This is a very low number
- Choice for the second polynomial: transform it in the hardware, or not
 - If the hardware does not transform it, that polynomial must arrive in step with the transformed first polynomial
- No block RAM for coefficients or for a twiddle table. Twiddle factors are computed as the data moves
- Each stage starts when its operands are ready and stores only that stage's alignment data, in registers and distributed memory
- No separate bit-reversal step
- Interface
 - Clock
 - Reset, active low
 - Two input polynomials and one output polynomial, each as coefficient beats marked valid or not
 - Backpressure: if the consumer does not accept the current output beat, that beat does not change and both inputs are held

- Measured Kyber results, degree 256, modulus 3329. The frequencies in these tables are for Artix-7 xc7a100tftg256-1L, a low-end FPGA. A better FPGA runs higher. On Virtex-7 xc7vx485tffg1157-1 the clock reached 206 MHz. Multiplications per second are the clock divided by beats per polynomial, with no backpressure.

Second polynomial not transformed in the hardware

Units per stage	NTT cycles	LUT	Flip-flops	LUTRAM	DSP	MHz	Million/s
1	128	4,482	4,139	567	22	155	1.21
2	64	7,210	6,806	916	36	152	2.38
4	32	13,320	12,668	1,613	72	148	4.63
8	16	25,196	24,142	2,827	144	143	8.93

Second polynomial transformed in the hardware

Units per stage	NTT cycles	LUT	Flip-flops	LUTRAM	DSP	MHz	Million/s
1	128	6,105	5,789	802	29	152	1.18
2	64	9,966	9,764	1,283	50	150	2.35
4	32	18,334	18,093	2,255	100	145	4.54
8	16	34,490	34,434	3,887	200	142	8.85

- Verification of the Kyber configuration
 - Products compared with an independent schoolbook multiplier in $Z_{3329}[x] / (x^{256} + 1)$, coefficients in order
 - Random polynomials, including back-to-back pairs and gaps between polynomials
 - A long run of random pairs, with idle time between them
 - Zero times zero, and a zero polynomial on either side
 - Multiply by the constant polynomial 1
 - Every coefficient at modulus minus one
 - A single nonzero coefficient, including the product that wraps through $x^{256} = -1$
 - The same pair multiplied twice
 - Both orders of the two polynomials
 - No backpressure, with the producer allowed to send every beat
 - Random backpressure
 - Backpressure on the first output beat, on the beat that finishes a polynomial, and a beat held for many cycles
 - Idle cycles only between polynomials
 - Reset while idle, reset in the middle of a polynomial, and reset while an output beat is held
 - Repeated at 1, 2, 4, and 8 butterfly units per stage, and for both choices of the second polynomial
 - Code coverage was collected

Deliverables

- RTL configured for the agreed degree, modulus, butterfly units per stage, and the choice of whether the second polynomial is transformed in the hardware
- Streaming interface described under Key features
- A bus wrapper on the streaming interface, when a bus is requested
- Verification, including the tests, the test report, and code coverage

Qryptic Inc. Contact: [email]